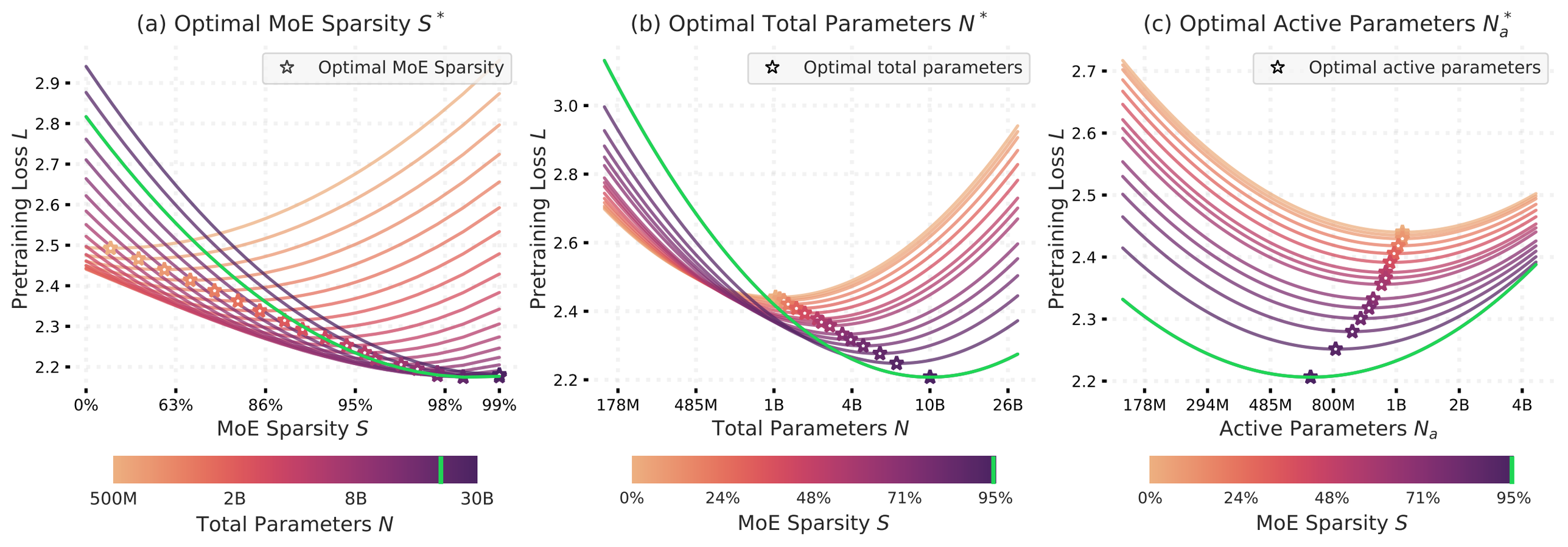
(a) Optimal MoE Sparsity S^*
Optimal MoE Sparsity
Pretraining Loss L
2.9
2.8
2.7
2.6
2.5
2.4
2.3
2.2
0%
63%
86%
95%
98%
99%
MoE Sparsity S
500M
2B
8B
30B
Total Parameters N
(b) Optimal Total Parameters N^*
Optimal total parameters
Pretraining Loss L
3.0
2.8
2.6
2.4
2.2
178M
485M
1B
4B
10B
26B
Total Parameters N
0%
24%
48%
71%
95%
MoE Sparsity S
(c) Optimal Active Parameters N_a^*
Optimal active parameters
Pretraining Loss L
2.7
2.6
2.5
2.4
2.3
2.2
178M
294M
485M
800M
1B
2B
4B
Active Parameters N_a
0%
24%
48%
71%
95%
MoE Sparsity S